

IMAGE QUALITY AND ARTIFACTS ON OPTICAL COHERENCE TOMOGRAPHY ANGIOGRAPHY

Comparison of Pathologic and Paired Fellow Eyes in 65 Patients With Unilateral Choroidal Melanoma Treated With Plaque Radiotherapy

EMIL A. T. SAY, MD, SANDOR FERENCZY, CRA, OCT-C, GEORGE N. MAGRATH, MD, MBA, WASIM A. SAMARA, MD, CHLOE T. L. KHOO, BS, CAROL L. SHIELDS, MD

Purpose: To study image quality and artifacts seen on optical coherence tomography angiography (OCTA).

Methods: Sixty-five consecutive patients with unilateral posterior uveal melanoma treated with plaque radiotherapy had OCTA during follow-up. Optical coherence tomography angiography was performed on both the affected and fellow eye. Signal strength and frequency of image artifacts on en face images were compared between affected and fellow eyes.

Results: A total of 130 eyes in 65 patients were analyzed, the mean age at time of OCTA was 55 years (median: 56, range: 12–81 years), and 39 (39/65, 60%) were female. Majority of tumors were located in the choroid (62/65, 95%) and extramacular (55/65, 85%). The mean distance to the foveola was 4 mm (median: 3, range: 0–18 mm) and optic nerve was 4 mm (median: 4, range: 0–16 mm). Optical coherence tomography angiography was performed at a mean 46 months after plaque radiotherapy. Most patients had a history of radiation maculopathy or papillopathy in the treated eye at the time of OCTA (46/65, 71%). Overall, 95 eyes (95/130, 73%) had at least one significant artifact on OCTA. The most common major artifacts were loss of focus (71/130, 55%), broad (>5 pixels width and >4 lines) blink lines (48/130, 37%), motion artifact (34/130, 26%), specular dot (33/130, 25%), and edge duplication (10/130, 8%). Statistically, eyes treated with plaque radiotherapy (affected vs. fellow eye) were more likely to have at least one major OCTA artifact (92 vs. 54%, $P < 0.001$) and, specifically, loss of focus was more frequent (78 vs. 31%, $P < 0.001$). Multivariate analysis found decreased visual acuity significantly associated with higher incidence of broad blink lines ($P = 0.0166$) and loss of signal ($P < 0.0001$), whereas male sex was associated with increased loss of signal ($P = 0.0015$), and distance to the foveola was related to edge duplication ($P = 0.0119$).

Conclusion: Image artifacts on OCTA are commonly encountered and appear to be more frequent in eyes with pathology and poor visual acuity. Recognition of these artifacts might help improve image interpretation and decision making.

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Optical coherence tomography (OCT) has changed the practice of ophthalmology since its introduction in 1991.¹ This technology has allowed in vivo cross-sectional analysis of the retina, and, subsequently, the anterior segment with remarkable anatomic tissue resolution. Technological advances have

lead to improved resolution down to 5 μm , faster scan speeds, real-time eye tracking to allow point-to-point comparison, multiimage averaging, and enhanced depth imaging OCT (EDI-OCT) for visualization of deep choroidal details.^{2,3} However, despite these innovations, there are artifacts of OCT that can cause

interpretation difficulty. Artifacts can occur with poor patient cooperation, media opacity, optical aberrations, and machine-related segmentation errors. These can lead to false-negative and false-positive scans, causing misinterpretation.⁴⁻¹¹ Artifacts are more often seen with older-generation time-domain OCT, but can still be seen in up to 69% of eyes imaged with newer-generation spectral-domain OCT and up to 9% of those on swept source OCT.⁴⁻⁵

Recently, OCT angiography (OCTA) has been introduced as a noninvasive imaging technique that allows visualization of both the superficial and deep capillary plexus, the latter of which cannot be seen with standard fundus fluorescein angiography.¹² This technology allows qualitative representation of blood flow in a limited area of the retina using computer-generated algorithms to determine flow in ocular tissue.^{12,13} We have published our initial experience with OCTA of eyes with choroidal melanoma treated with plaque radiotherapy and documented microcapillary ischemia, even in eyes without visible radiation retinopathy.^{14,15} In this analysis, we specifically evaluate OCTA image quality and image artifacts, with emphasis on the types and frequency of artifacts, and factors associated with reduction of image quality and prevalence of image artifacts.

Materials and Methods

A retrospective chart review of all patients diagnosed with posterior uveal melanoma treated with plaque radiotherapy who had OCTA between February 2015 and April 2015 were included in the study. This study adhered to the tenets of the Declaration of Helsinki and institutional review board of Wills Eye Hospital approval was obtained. Informed consent for use of the OCTA device was obtained from all subjects after explanation of the nature of the study. Data gathered included patient demographics, preoperative clinical features, postoperative clinical features at the time of OCTA, and OCTA parameters and findings. In addition, central macular thickness (CMT)

Table 1. Image Quality and Artifacts on Optical Coherence Tomography Angiography (OCTA)

	n = 65, N (%)
Age, mean (median, range), years	55 (56, 12–81)
Race	
Caucasian	63 (98)
Hispanic	2 (2)
Sex	
Male	26 (40)
Female	39 (60)
Past medical history	
Diabetes	4 (4)
Hypertension	25 (38)
Involved eye	
Right	33 (51)
Left	32 (49)
Tumor quadrant	
Superior	12 (18)
Nasal	15 (23)
Inferior	14 (22)
Temporal	14 (22)
Macula	10 (15)
Mean distance to foveola (median, range)	4 (3, 0–18)
Mean distance to optic nerve (median, range)	4 (3, 0–16)
Mean interval from plaque (median, range), months	45 (26, 3–216)
Radiation complications:	
Radiation maculopathy (any time)	34 (52)
Radiation maculopathy at time of OCTA	26 (40)
Radiation papillopathy (any time)	10 (15)
Radiation papillopathy at time of OCTA	8 (12)
No radiation complications at time of OCTA	19 (29)

Clinical features.

on standard cross-sectional OCT at the time of study was also recorded for each eye. All patients in this cohort have been previously evaluated and reported in a separate publication regarding OCTA microischemia after plaque radiotherapy of uveal melanoma.¹⁴ In this report, the analysis was specifically focused on image quality and artifacts.

Patient demographics include age, race, sex, and past medical history (diabetes mellitus, hypertension). Visual acuity at the time of OCTA study was documented, as well as clinical features of radiation complications after plaque radiotherapy. Clinical ocular and tumor features included tumor size, location, distance to the foveola, and distance to the optic nerve.

Image Acquisition Protocol

The Optovue RTVue XR AVANTI, version 2014.2.0.13 (Optovue Inc, Fremont, CA) was used to obtain all study images in this analysis. It uses an

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C. L. Shields has had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Reprint requests: Carol L. Shields, MD, Ocular Oncology Service, 840 Walnut Street, Suite 1440, Philadelphia, PA 19107; e-mail: carolshields@gmail.com

Table 2. Image Quality and Artifacts Seen on Optical Coherence Tomography Angiography (OCTA)

	Treated Eyes (n = 65), N (%)	Control Eyes (n = 65), N (%)	P	Total (n = 130), N (%)
Mean Snellen visual acuity (median, range)*	20/100 (20/60, 20/20-NLP)	20/25 (20/20, 20/20–20/40)	<0.001	20/50 (20/30, 20/20-NLP)
Mean CMT (median, range), microns*	300 (277, 132–652)	271 (268, 235–339)	0.0499	285 (271, 132–652)
Mean OCTA signal strength (median, range)*	51 (52, 17–79)	68 (68, 43–86)	<0.001	60 (62, 17–86)
OCTA artifacts†				
Specular dot	9 (14)	24 (37)	0.0043	33 (25)
Motion artifact	19 (29)	15 (23)	0.5498	34 (26)
Edge duplication	8 (12)	2 (3)	0.0958	10 (8)
Blink lines (any)	61 (93)	57 (88)	0.3642	118 (91)
Blink lines, broad	28 (43)	20 (31)	0.2031	48 (37)
Media opacity	5 (8)	1 (2)	0.2078	6 (5)
Loss of signal	4 (6)	0 (0)	0.1192	4 (3)
Loss of focus	51 (78)	20 (31)	<0.001	71 (55)
No major OCTA artifact‡	5 (8)	30 (46)	<0.001	35 (27)

Summary of OCTA image quality and artifacts. The *p*-values in bold and italics were considered statistically significant.

*Paired Student's *t*-test.

†Fisher's exact test.

‡Major OCTA artifacts include motion artifact, edge duplication, broad (>5 pixels and >4 in number) blink lines, media opacity, loss of signal, and loss of focus.

CMT, central macular thickness; OCTA, optical coherence tomography angiography.

840-nm wavelength laser with a bandwidth of 45 nm, 3 mm scan depth, and tissue resolution of 5 μ m. This technology can acquire 70,000 A-scans per second and up to 4,096 A-scans per B-scan. All patients required a standard retinal map analysis protocol for volume scans. Central macular thickness (CMT) was recorded for analysis. The same machine was also used for OCTA image acquisition. The scanning algorithm began with two B-scans captured at each fixed position before the next sampling location and there were two orthogonal OCTA volume scans (one horizontal and one vertical) used to minimize motion artifacts and fixation changes. Split-spectrum amplitude-decorrelation angiography (SSADA) algorithm was used to create OCTA en face images. The 3 mm scans were used for analysis to magnify vascular detail. Image segmentation was automatically performed, using the OCT camera's built-in software (ReVue, version 2014.2.0.65). The superficial capillary plexus en face image was automatically segmented with an inner boundary at 3 μ m beneath the internal limiting membrane (ILM) and an outer boundary set at 15 μ m beneath the inner plexiform layer (IPL), while the deep capillary plexus en face image had its inner boundary at 15 μ m beneath the IPL and an outer boundary at 70 μ m beneath the IPL. We selectively included only the first OCTA scan set obtained from each patient for analysis, regardless of quality and signal strength. Specifically, the first scan will always be the normal fellow eye, and the photographer immediately moves to the diseased study eye for the next (second) scan. All

OCTA images were taken by a single photographer (SF) and further OCTA images may be taken but were not included in the analysis. Optical coherence tomography angiography parameters included signal strength of each scan, segmentation errors, and specific artifact.

Statistical Analysis

All recorded data and statistical analyses were performed using Microsoft Excel 2011. Fisher's exact test was used to compare categorical variables between treated eyes and the fellow untreated control eyes, while paired Student's *t*-test was used to compare continuous data such as signal strength and visual acuity. Linear regression analysis was used to correlate the frequency of OCTA artifacts with patient demographics and clinical variables. The Fisher exact test was used to verify these relationships when applicable. Factors that were significant on univariate analysis (*P* < 0.05) were then analyzed in a multivariate linear regression model to determine whether these variables were independently associated. For this analysis, the eye that received plaque radiotherapy was referred to as the treated eye, while the fellow, nontreated eye was denoted as the control eye.

Results

There were 130 eyes in 65 subjects included in the study and patient demographic features at the time of study are listed in Table 1. The mean patient age was

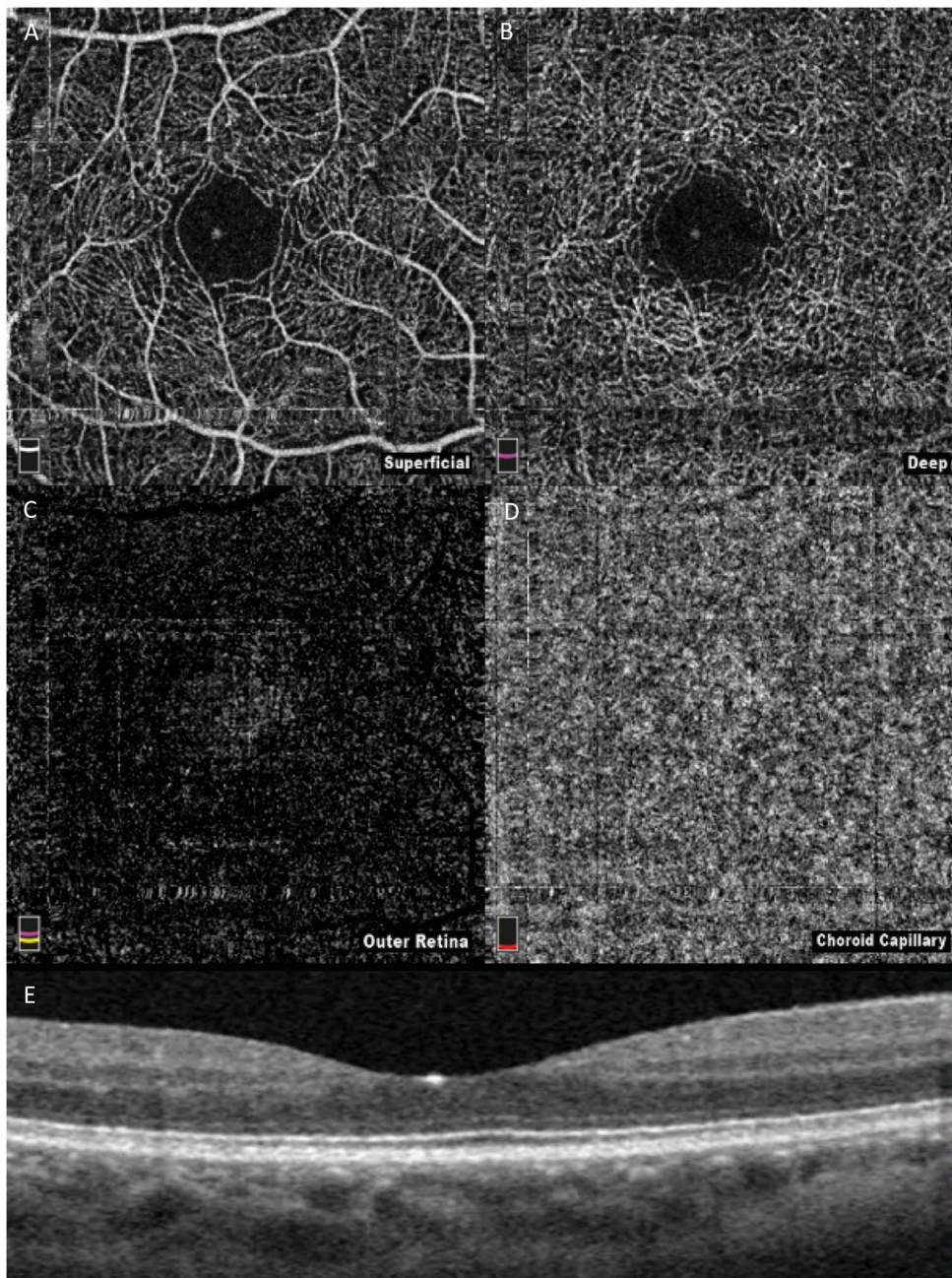


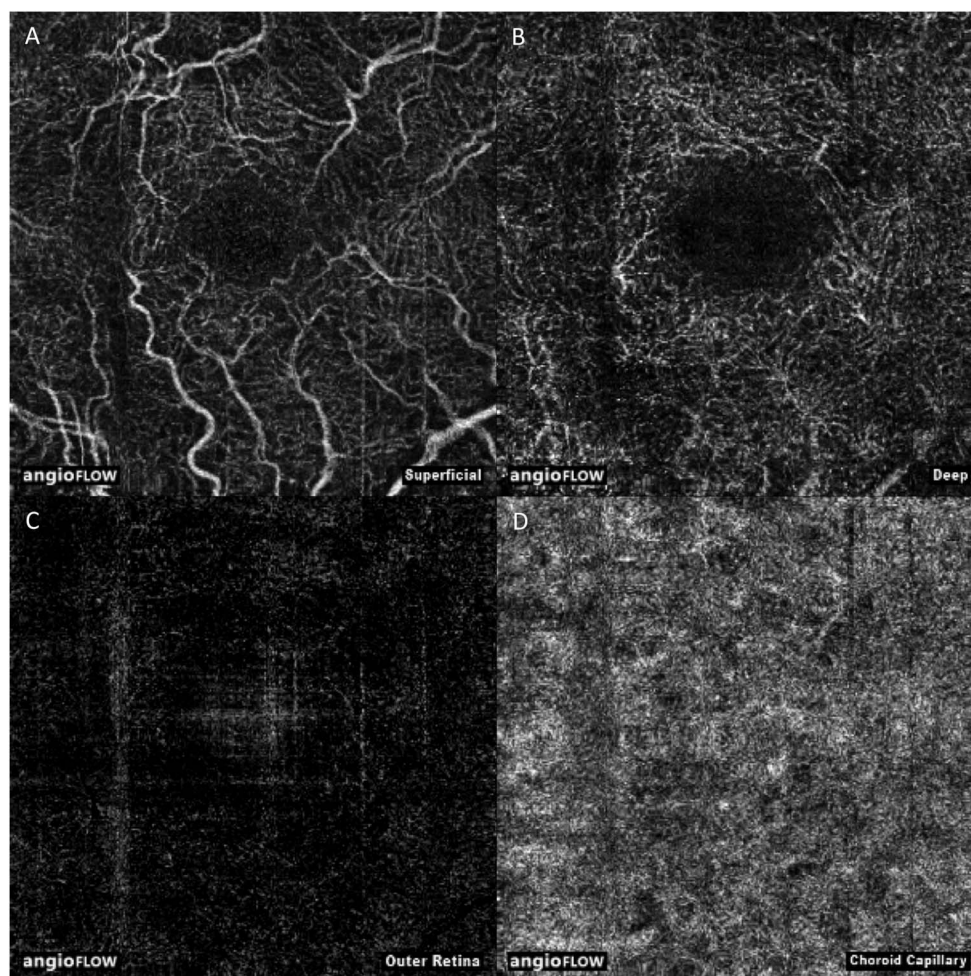
Fig. 1. Specular dot artifact in the normal fellow eye of a patient. On optical coherence tomography angiography, the superficial (A) and deep (B) plexus en face image show the bright specular reflection in the central foveola representing a perfect tangential alignment of the scanning laser interferometer with the foveola (“umbo”). This reflection is absent on deep scans such as the outer retina (C) and choriocapillaries (D). Its correlate on the corresponding center line scan of the en face image (E) is a bright hyperreflective lesion in the trough of the foveal depression.

55 years (median 56, range: 12–81 years) and most were female (39/65, 60%) and Caucasian (63/65, 98%). No patient demonstrated clinical evidence of diabetic or hypertensive retinopathy although one patient with diabetes mellitus showed subclinical, occult microaneurysms on OCTA consistent with non-proliferative diabetic retinopathy. Overall, in the irradiated eyes ($n = 65$), there were 34 eyes (34/65, 52%) with clinical evidence of radiation maculopathy, 29 (29/65, 45%) with OCT evidence of radiation-related macular edema, and 10 (10/65, 15%) with radiation

papillopathy. There were 19 eyes (19/65, 29%) without evidence of radiation complications. In the fellow control eyes ($n = 65$), there was no visible retinal vascular pathology in any case.

Optical coherence tomography angiography was obtained at mean 46 months (median: 26 months, range: 4–216 months) after plaque radiotherapy. Mean Snellen visual acuity was 20/100 (median: 20/60, range: 20/20–NLP) and 20/25 (median: 20/20, range: 20/20–20/40) for treated and control eyes, respectively. Mean CMT was 300 μm (median: 277 μm ,

Fig. 2. Motion artifact in a patient with extramacular choroidal melanoma. This artifact is represented as vessel doubling that is best seen on scans through the inner retina, through the superficial (A) and deep (B) plexus, because of the prominent network of vessels. In contrast, the absence of any normal vascularity in the outer retina (C), while the lobular network of vessels in the choroid makes this artifact less pronounced on scans through the choriocapillaries (D).



range: 132–652 μm) in the treated eye and 271 μm (median: 268 μm , range: 235–339 μm) in the control eye. On OCTA, mean signal strength was 51 (median: 52, 17–79) in the treated eyes compared with 68 (median: 68, range: 43–86) in control eyes, and the difference was statistically significant ($P < 0.001$). The types of image artifacts are discussed below.

Optical Coherence Tomography Angiography Image Artifacts

There were several artifacts assessed in this report, some of which have been previously described.^{16,17} These artifacts include specular dot, motion artifact, edge duplication, blink lines, media opacity, loss of signal, motion artifact, and loss of focus (Table 2). The specular dot is a bright, reflective artifact toward the foveolar center. It appears brightly reflective on en face images, visible at the level of the superficial and deep plexus, and in this cohort, was found only in patients with normal foveal contour (Figure 1). This

artifact likely represented the physiologic umbo anatomically and the foveal light reflex clinically. The specular dot correlated with focal hyperreflectivity at the inner foveolar surface on OCT.¹⁸ Specular dot was significantly more common in control eyes (treated vs. control eyes) (37 vs. 14%, $P = 0.0043$).

Motion artifact is mostly patient dependent but may also occur with software motion correction technology. It occurs when significant transverse eye motion occurs from microsaccades and fixation loss.¹⁶ This appears as doubling of the retinal vessels or a white line with lateral displacement of retinal vessels (Figure 2). Although the Optovue RTVue XR AVANTI uses split-spectrum amplitude-decorrelation angiography to minimize effects of axial motion and improve signal-to-noise ratio, reduction of motion artifacts from transverse movement is dependent on Optovue's motion correction technology (MCT). In this analysis, motion artifact was similar in both groups (29 vs. 23%, $P = 0.5498$).

Edge duplication refers to linear white streaks with varying width at the edge of the en face image

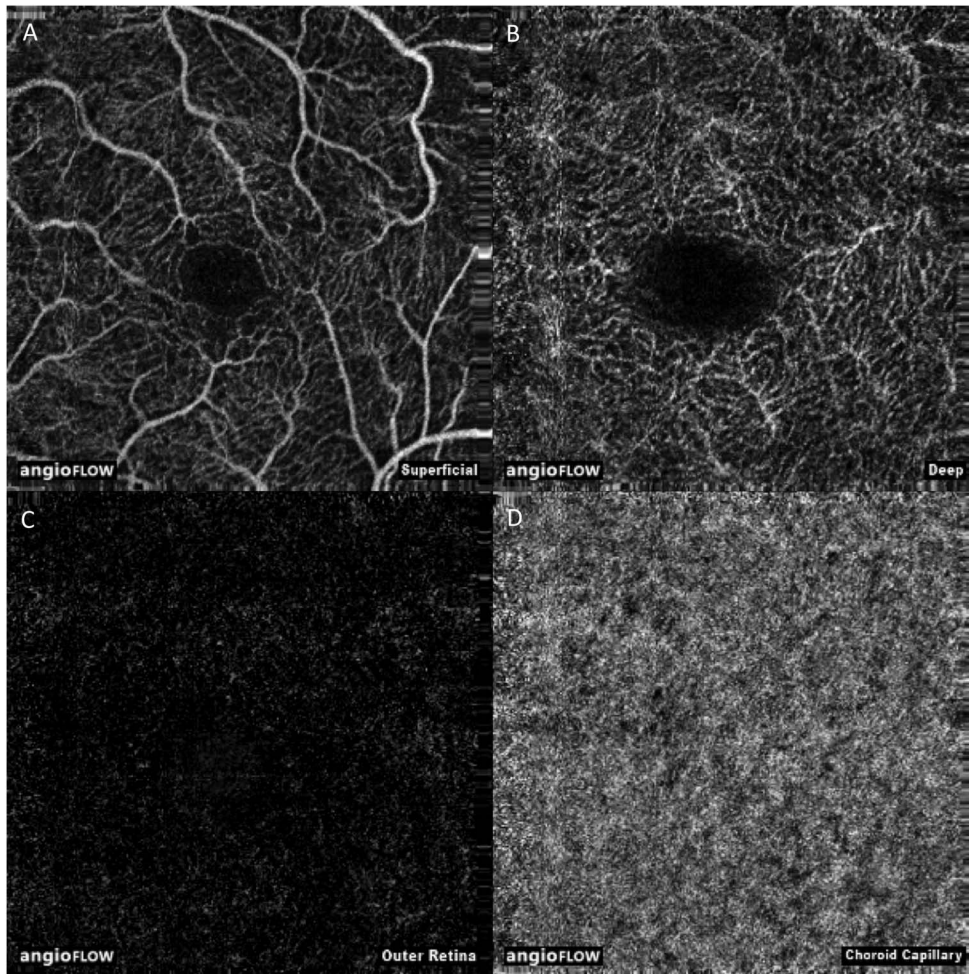


Fig. 3. Edge duplication artifact in an eye with extramacular choroidal melanoma and treated radiation maculopathy. The edge duplication artifact refers to the linear streaks at the edge of an en face image representing intermittent loss and gain of signal because of microsaccades and small angle change in focus. It is always visible en face cuts, including the superficial (A) and deep (B) plexus, outer retina (C), and choriocapillaries (D).

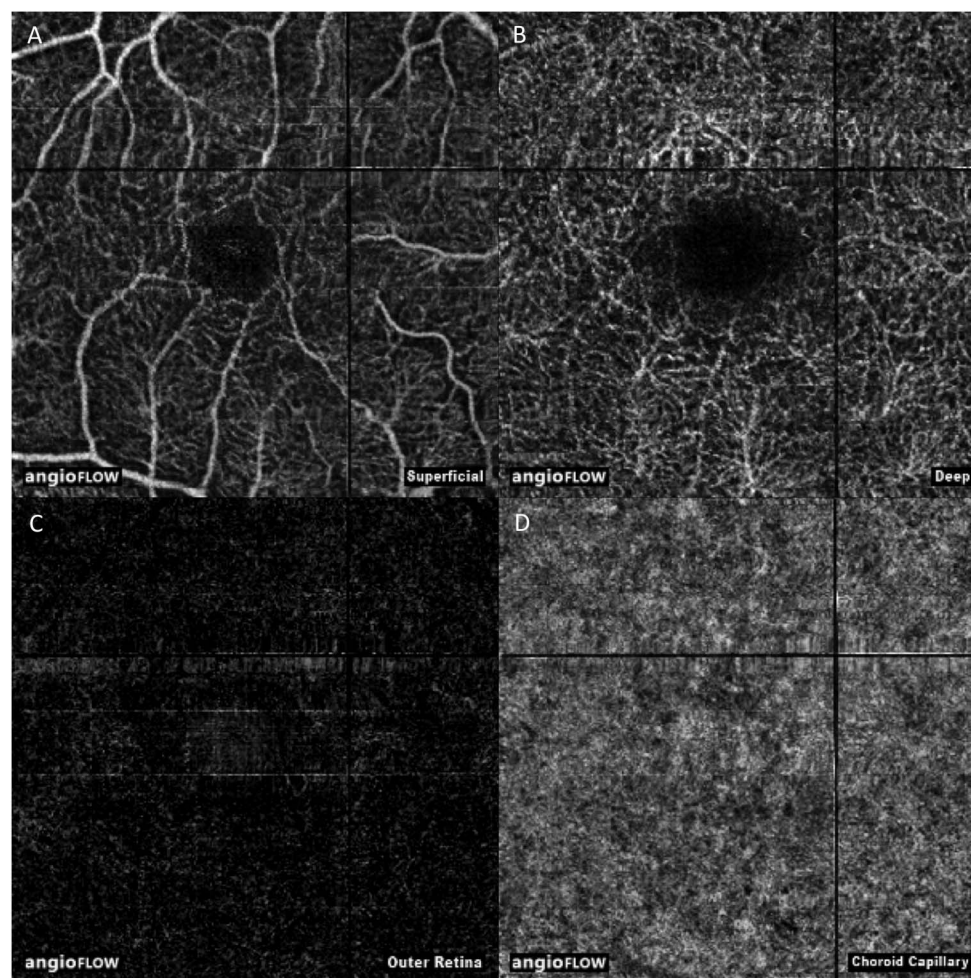
(Figure 3). This is likely similar to the stretch artifact described by Spaide and considered to be a byproduct of the sequential scan sets and postcapture motion correction technology applied to the final OCTA en face image.¹⁶ When sequential, orthogonal OCT volume scans are used to produce OCTA en face images, motion correction technology must estimate the eye movement for every corresponding A-scan in the two sets of volume data. The A-scan movement estimates can introduce motion artifacts at the most time distant B-scans, where the distortion from motion can be the most evident, producing artifacts such as edge duplication, where false flow is detected because of misalignment of A-scans. In our analysis, edge duplication artifact was nonsignificantly more common in treated eyes (12 vs. 3%, $P = 0.0958$).

Blink lines were the most frequently seen OCTA artifact and appears as an end-to-end dark line of varying width and through all depths of the en face scans.¹⁷ The length of blink lines is characteristically the length of the B-scan and is not interrupted

(Figure 4). However, the width varies depending on the duration of the blink. We found similar rate of blink lines in treated and control eyes (93 vs. 88%, $P = 0.3642$). We considered an image with more than 4 blink lines each comprised of a pixel width of 5 pixels as a broad blink defect and used a standard image with this predetermined minimum pixel width to determine the frequency of broad blink lines (Figure 3). In this analysis, there was similar frequency of broad blink lines in treated eyes and control eyes (43 vs. 31%, $P = 0.2031$).

Media opacities on OCTA are full thickness and seen at every depth of segmentation (Figure 5). When compared to other hyporeflective image artifacts, they appear with softer and fuzzier margins, rather than the abrupt, distinct margins caused by segmentation errors or loss of signal. Also, media opacities typically have a clinical correlate, such as vitreous blood, asteroid hyalosis, or vitreous opacification.¹⁶ Our analysis showed similar low frequency of media opacity in both groups (8 vs. 2%, $P = 0.2078$).

Fig. 4. Blink line artifact in an eye with extramacular choroidal melanoma and no history of radiation maculopathy. Blink lines appear as end-to-end horizontal or vertical lines that appear in all en face images, including the superficial (A) and deep (B) plexus, outer retina (C), and choriocapillaries (D). The image represented here has one horizontal and one vertical blink line that measured 5 pixels in width and was used as the standard image for a broad blink line.



Loss of signal is patient dependent, occurs primarily in diseased eyes, and appears as abrupt loss of OCT signal secondary to segmentation error because of abrupt elevation (tumor, retinal edema, or thickening) or depression (loss of retinal tissue) of the retinal topography (Figure 6). The correlate on OCT B-scan is an abrupt cut edge or mirror artifact when the plane of focus extends beyond the machine's scan depth.^{7,9} Loss of signal was similar in both groups (6 vs. 0%, $P = 0.1192$) and notably absent in control eyes.

Loss of focus is operator dependent rather than patient dependent and appears as a localized area of signal loss on the en face image (Figure 7). The Optovue RTVue XR AVANTI uses scanning laser interferometry to obtain serial B-scans. The laser has a set focal length and a plane of focus outside the predetermined zero-delay point and can lead to artificial signal loss and, in turn, interpretation as loss of flow and nonperfusion. This can be artificially reproduced by deliberately changing the plane of focus by a few diopters (Figure 8). Loss of focus artifact was significantly more common in treated eyes (78 vs. 31%, $P < 0.001$).

There is significant heterogeneity in the occurrence of OCTA image artifacts. To determine image quality, we also subdivided eyes into those with no major artifacts (motion artifact, edge duplication, broad blink lines, media opacity, loss of signal, and loss of focus). We found lack of artifact significantly more common in control eyes (46 vs. 8%, $P < 0.001$) (Table 2).

Clinical Features Associated With Image Artifacts

In addition to comparison of treated versus control eyes, subgroup analysis was performed to determine clinical factors associated with specific artifacts (Tables 3 and 4). On univariate analysis, broad blink lines occurred significantly more commonly in patients with poor visual acuity ($r = 0.2977$, $P = 0.0288$). Edge duplication was significantly related to tumors further away from the foveola ($r = 0.3867$, $P = 0.0039$). Loss of signal occurred significantly more commonly in males ($r = 0.3411$, $P = 0.0116$) and patients with poor visual acuity ($r = 0.5868$, $P < 0.0001$), whereas loss of focus was significantly and inversely related to low signal

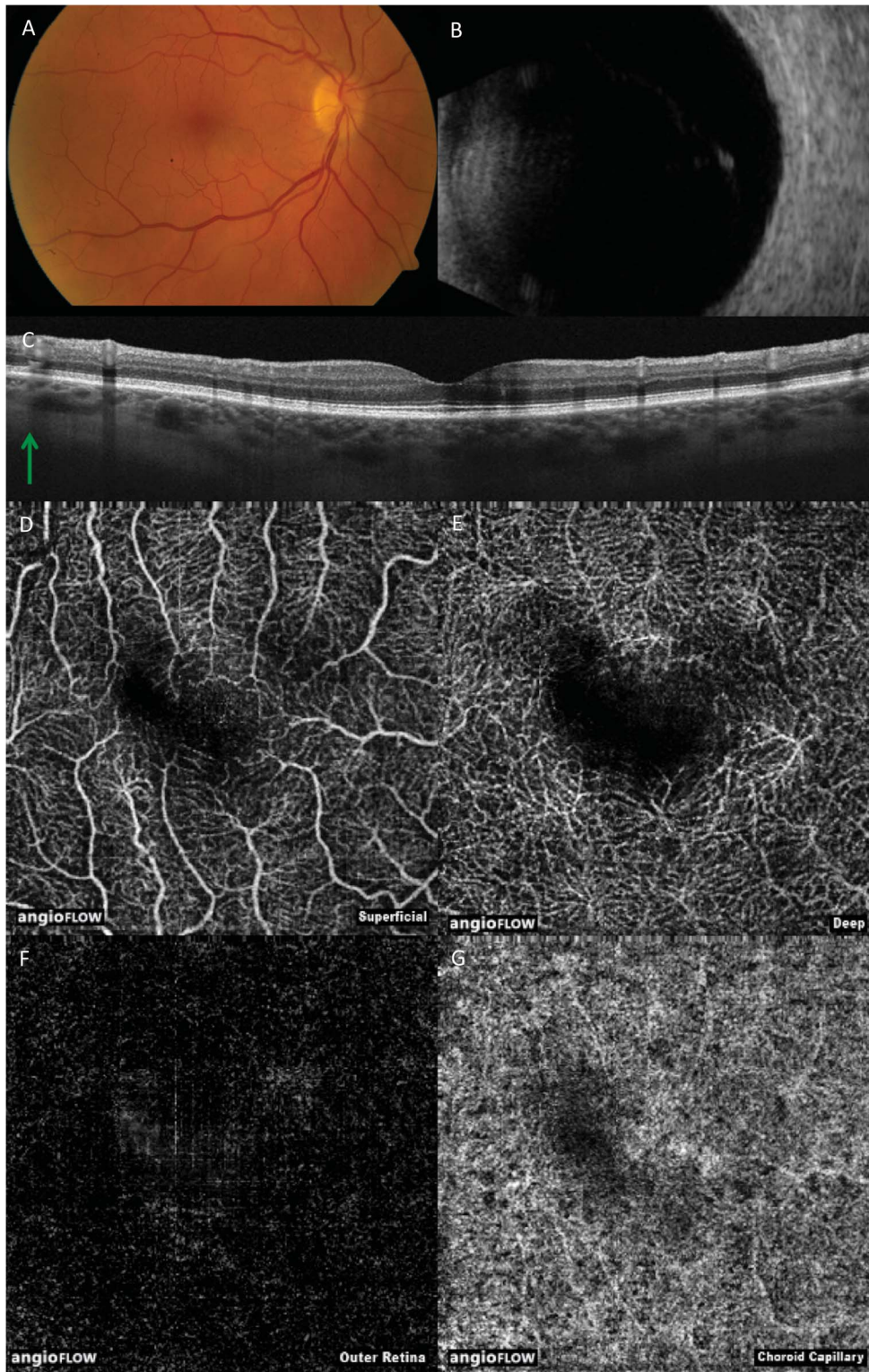
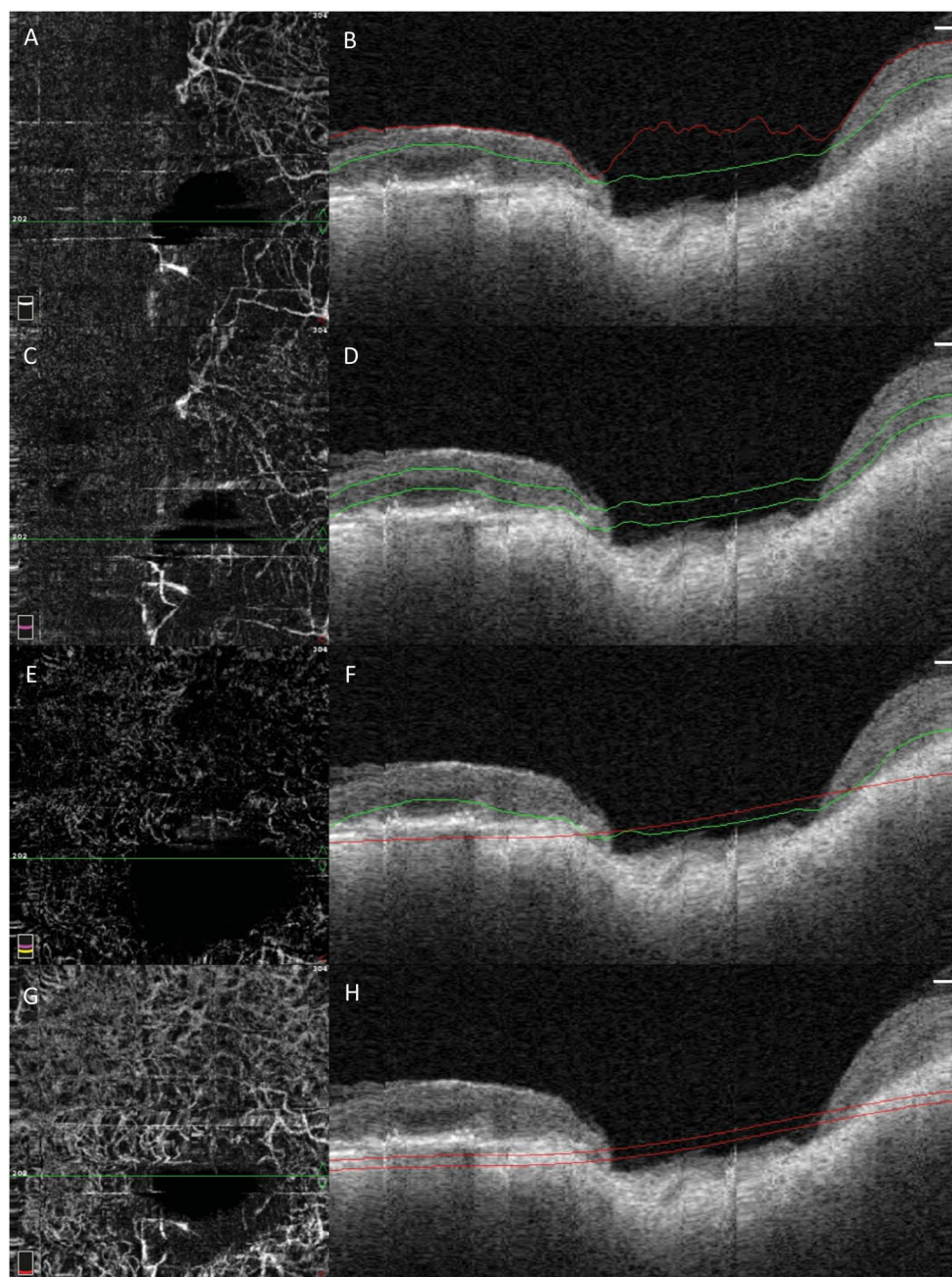


Fig. 5. Media opacity artifact in the right eye of a patient with choroidal melanoma in her opposite eye. Fundus photograph (A) focused on the macula appears normal; however, a small echodense vitreous floater can be visualized by ultrasonography (B). The optical coherence tomography vertical cross-line scan (C) showed shadowing from the vitreous floater in the superior foveola and adjacent parafoveal region. By optical coherence tomography angiography, the vitreous floater can also be visualized in the superotemporal juxtafoveal region, appearing as a dark shadow causing artificial enlargement of the superficial (D) and deep (E) plexus. Shadowing becomes less apparent in the outer retina (F) because of its avascularity, but again becomes visible in at the level of the choriocapillaries (G).

strength ($r = -0.2740$, $P = 0.0451$). Using a multivariate linear regression model, poor visual acuity significantly predicted broad blink lines ($\beta = 0.2769$, $P = 0.0166$) and loss of signal artifact ($\beta = 0.2260$, $P < 0.0001$),

tumor distance to foveola significantly predicted edge duplication artifact ($\beta = 0.0357$, $P = 0.0119$), and male patients were more likely to have loss of signal artifacts ($\beta = 0.1987$, $P = 0.0015$) independent of other

Fig. 6. Loss of signal artifact in a patient with extramacular choroidal melanoma in the right eye and severe macular ischemia with retinal thinning after treatment. On optical coherence tomography angiography (OCTA), each en face image had a geographic black region in the parafoveal region arising from errors in the automated segmentation through an area with loss of retinal tissue that can be detected in each corresponding optical coherence tomography (OCT) b-scan image. This can be visualized in the OCTA en face (A, C, E, and G) and Verified as a segmentation error in the corresponding OCT b-scan (B, D, F, and H) seen At the level of the superficial plexus (A and B), deep plexus (C and D), outer retina (E and F), and choriocapillaries (G and H). The loss of signal artifact in this case can be distinguished from areas of non-perfusion at the level of the superficial (A) and deep (C) plexus because of its distinctly darker appearance, as compared to the ischemic temporal retina.



predictors. Signal strength was not a significant predictor of edge duplication in univariate analysis ($P = 0.0963$); however, it was significant in the multivariate analysis ($P = 0.0483$), indicating a small and likely insignificant relationship.

Discussion

Artifacts are inherent to all imaging devices. Some are generally observed with several imaging technologies,

such as media opacity, while others are exclusive to a specific device. In traditional film or digital fundus photography, small immovable translucent opacities could be related to dust on the lens or camera sensor. Ultra-widefield scanning laser ophthalmoscope (Optos, Dunfermline, Scotland, United Kingdom) often reveals eyelash artifact, warping, changes in perspective, and artificial color artifact that appear different from traditional white light fundus photography, while near-infrared reflectance scanning laser ophthalmoscope images (Heidelberg, Franklin, MA)

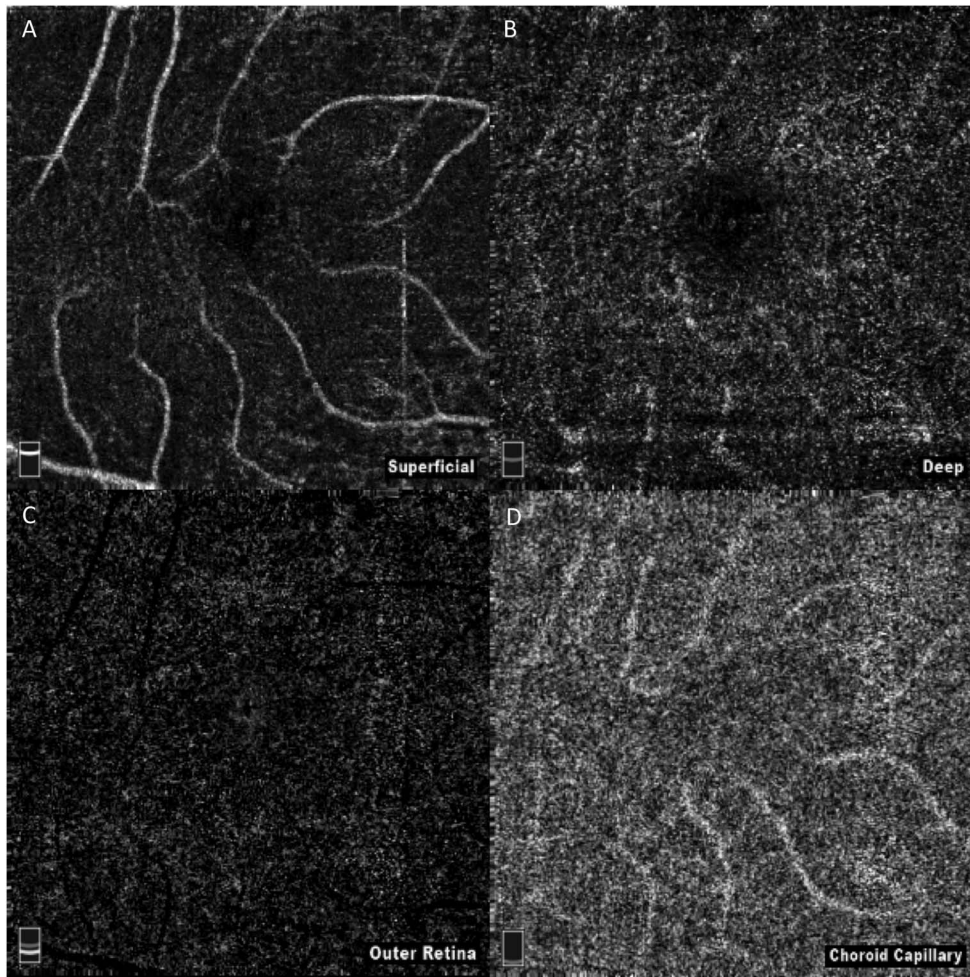


Fig. 7. A representative example of a normal patient with loss of focus artifact. In this case, loss of focus was artificially created by deliberately defocusing the image by +5.00 diopters on the z-axis. This created an artificial zone of hyporeflectivity in areas with finer vasculature such as the capillaries of the superficial (A) and deep (B) plexus. This is less visible on the outer retina (C) and choriocapillaries (D). There is an incidental projection artifact of the superficial vessels seen in both the outer retina (dark vascular network) and choriocapillaries (bright vascular network).

often reveal hyperreflective artifact caused by light scatter in pseudophakic patients, termed “ghost maculopathy.”¹⁹

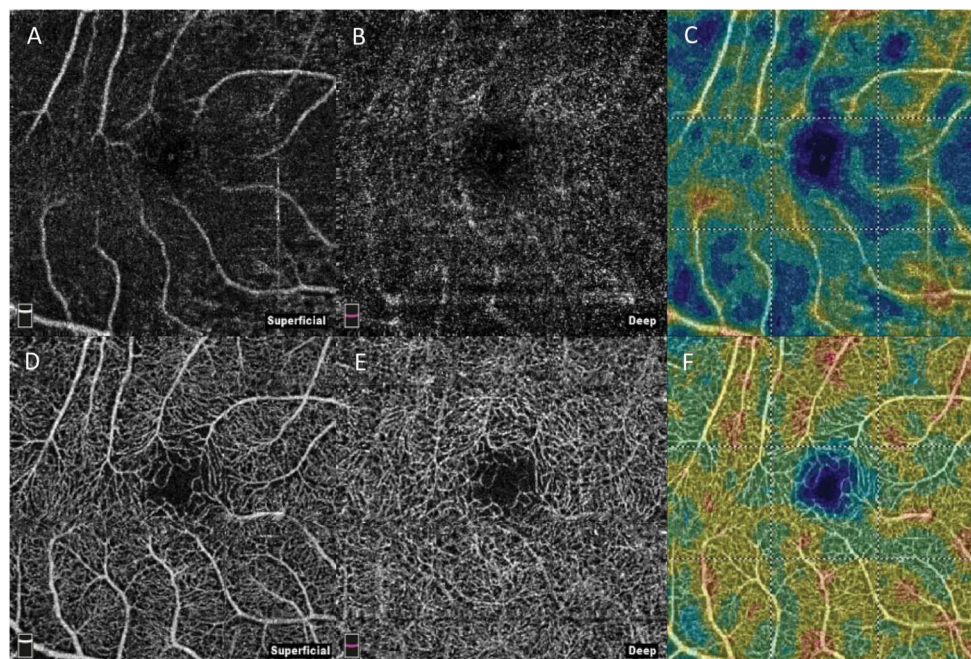
Standard OCT artifacts can be machine or patient related. Some machine-related artifacts include segmentation errors, out of register, cut edge, double fovea, and mirror image artifacts, while patient-related artifacts include full thickness waves signifying patient movement.^{4–11} Software and hardware applications can be designed to minimize these artifacts. Regarding OCT, upgraded features like high-speed eye tracking and faster scan speeds have reduced some artifacts.⁴ Furthermore, it is known that OCT artifacts are more frequent in diseased eyes.^{20–22}

Optical coherence tomography angiography has issues regarding image quality and artifacts. In this study, 73% of all 130 eyes imaged had at least one major artifact although control eyes showed better image quality (higher signal strength) and fewer artifacts overall.^{20–22} The most common artifact was the blink line, followed by loss of focus and motion

artifact. The impact of these artifacts relates to loss of data and reduction in image analysis (Figure 7). Blink lines are interpreted as loss of flow and appear as a dark horizontal or vertical line on all en face slabs. When analyzed for capillary density or foveal avascular zone size (FAZ), blink lines are likely to be misinterpreted as areas of low flow, or even nonperfusion, artificially reducing capillary density and enlarging FAZ. In contrast, motion artifact, edge duplication, and loss of focus may be erroneously interpreted as increased capillary flow and vascularity. Hence, the interpreter should exercise caution when taking measurements from the en face image, as poor image quality could either positively or negatively affect analysis.^{23,24}

The high occurrence of a central specular dot in normal eyes is expected because the foveal reflex occurs as a reflectance of the internal limiting membrane at the foveola and represents a normal contour.^{19,25,26} It should be noted that all 9 eyes in the treated group with specular dot artifact had normal foveal contour.

Fig. 8. Montage of the same patient in Figure 7 accompanied with capillary density maps. The same case as Figure 8 with intentional loss of focus showing loss of capillaries in the superficial (A) and deep (B) plexus represented as blue areas on the capillary density map (C). Correcting the image to plano shows the intact superficial (D) and deep plexus (E), with a true capillary density map (F). In this case, there was a significant reduction in capillary density of the en face image from 42% while defocused to +5.00 diopters to the true capillary density of 56% when the image was corrected to plano.



There are other artifacts not quantified in this study. These include projection artifacts and segmentation errors from more superficial vasculature “ghosting” on deeper structures^{16,17} These artifacts are inherent to the machine and some may be partially corrected using proprietary software enhancements. We realize that these are important artifacts in any setting, but segmentation errors can be manually corrected by adjusting each segmentation line of each B-scan image without retaking an OCTA image and hence is not entirely patient or operator dependent. Further, some of the artifacts described here and elsewhere may also be related to segmentation errors that are seen only on the B-scan with varied representations on the en face image, such as loss of signal. We estimate that in pathologic eyes, there will be a high number of segmentation errors and this has also been previously described with traditional OCT retinal maps.^{20–22} Projection artifacts, however, are inherent to current OCTA technology and are important artifacts as these can artificially increase density. It can be also quite difficult to tell for certain that each capillary of within the deep plexus is a projection of the superficial plexus, and because of the difficulty, imprecision, and inconsistent method of quantifying projection artifacts, we have elected not to include this in our analysis until improvements in technology are available.

Multivariate analysis of OCTA artifacts revealed that poor visual acuity was related to broad blink lines ($P = 0.0166$) and loss of signal ($P < 0.001$), eyes with melanoma farther from the foveola demonstrated edge

duplication ($P = 0.0119$), and males showed greater loss of signal artifact ($P = 0.0015$). The negative relationship of poor visual acuity to broad blink lines and loss of signal could indicate macular pathology and irregularities of the retinal contour, as many affected eyes demonstrated macular tumors, submacular fluid, or macular thickening from cystoid edema that caused reduction in focus and central foveolar alignment. Additionally, other factors like radiation-related dry eye could play a role in fixation.

The effects of tumor proximity to the foveola on edge duplication can be related to multiple factors. In general, tumors in this study were a mean 4 mm away from the foveola, indicating that in a significant subset of eyes, there is tumor present at the edge of the 3 mm × 3 mm scan area and could lead to segmentation errors across multiple B-scans and minute motion-related signal loss at the edges of where the tumor is located. Signal strength was a machine-generated number that reflects image quality. In standard OCT, loss of signal strength negatively affects thickness measurements and increased variability.^{5,27–29} In the present study, signal strength appeared to predict only loss of focus artifact. Both loss of focus and signal strength are both operator dependent, requiring technique to avoid. This cohort was consecutively recruited from the initial OCTA experience after this technology was commercially available. We anticipate that operator-dependent artifacts will lessen with experience.

There are a few limitations of this study. First, we analyzed major image artifacts but there could have

Table 3. Image Quality and Artifacts Seen on Optical Coherence Tomography Angiography (OCTA)

Predictors	Pearson r Coefficient (P)					
	Broad Blink	Media Opacity	Edge Duplication	Loss of Signal	Loss of Focus	Motion Artifact
Age	0.1175 (0.3974)	0.1134 (0.4143)	0.0376 (0.7871)	−0.1054 (0.4479)	0.0979 (0.4813)	0.0341 (0.8064)
Sex	−0.1044 (0.4523)	−0.0366 (0.7929)	0.0786 (0.5722)	0.3411 (0.0116)	0.0451 (0.7463)	0.1333 (0.3368)
Visual acuity	0.2977 (0.0288)	−0.1031 (0.4580)	−0.1038 (0.4527)	0.5868 (<0.0001)	−0.029 (0.8351)	−0.1145 (0.4095)
Distance to foveola	−0.0792 (0.5692)	0.0301 (0.8284)	0.3867 (0.0039)	−0.1840 (0.1829)	0.1549 (0.2633)	−0.0466 (0.7378)
Central macular thickness	−0.0551 (0.6924)	−0.0297 (0.8311)	0.2608 (0.0568)	−0.0854 (0.5392)	0.0967 (0.4869)	−0.0434 (0.7554)
Signal strength	−0.1145 (0.4099)	−0.0260 (0.8519)	0.2287 (0.0963)	−0.1820 (0.1878)	−0.2740 (0.0451)	−0.05293 (0.7039)

Univariate analysis of clinical features associated with major OCTA artifacts. The *p*-values in bold and italics were considered statistically significant.

been minor artifacts that were overlooked and we excluded analysis of the incidence of segmentation errors on individual B-scans and projection artifacts. We realize these are important in determining image quality; however, the primary focus of this report was to analyze the real-world occurrence of artifacts in a clinical setting. The interpretation of OCTA images are mostly done through the en face mode and there are a number of individual B-scans underlying each single OCTA image, so it was our choice to evaluate the final relevant product rather than all underlying components. Individual B-scans per OCTA image are of lower quality compared with similar B-scans created using the cross line or raster protocol because of the rapid split-spectrum amplitude-decorrelation angiography algorithm to achieve faster scans and minimize axial movement at the cost of lower signal-to-noise ratio.^{13,16} The second limitation relates to the small sample size and monocular involvement of all patients. The smaller number of patients in our series limits the ability to achieve statistical significance between some groups, especially if the difference between these groups is small. The monocular status of retinal pathology is a unique feature of our population and has both advantages and disadvantages compared with other more common retinal diseases with predominantly binocular involvement such as age-related macular degeneration, diabetic retinopathy, and other retinal vascular diseases. The monocular pathology in our population allowed us to singularly test the ocular pathology itself as a factor related to image quality and the frequency of artifacts, and limited the involvement of bias from patient factors because study controls in this cohort was the paired fellow eye of the same patient. However, all patients in our series had previous ocular surgery in at least one eye, which could have predisposed to blinking from dry eye, leading to blink artifact in both eyes. Future studies comparing young, healthy individuals who have not had previous ocular surgery and diseased eyes may be necessary to determine if prior ocular surgery or other factors might affect the occurrence of these aberrations. Lastly, our study is limited to the first (normal fellow eye) and second (diseased study eye) OCTA scans that the patient had throughout their visits in our clinic, which also can be associated with positive and negative factors. It is known that similar to other imaging devices, there is a learning curve present with both patient and photographer, and it is likely that images acquired after the first ever scan will be of better quality than the first ever scan. We attempted to reduce the effects of the learning curve on image quality and prevent masking of image quality of diseased eyes because of this learning curve. What

Table 4. Image Quality and Artifacts Seen on Optical Coherence Tomography Angiography (OCTA)

Predictors	Unstandardized β (P)					
	Broad Blink	Media Opacity	Edge Duplication	Loss of Signal	Loss of Focus	Motion Artifact
Age	0.0093 (0.0732)	0.0016 (0.5218)	0.0761 (0.3207)	-0.0009 (0.6807)	0.0002 (0.9590)	-0.0024 (0.6494)
Sex	-0.1917 (0.1830)	-0.0350 (0.6218)	0.0975 (0.3135)	0.1987 (0.0015)	0.0364 (0.7591)	0.1262 (0.3836)
Visual acuity	0.2769 (0.0166)	-0.0288 (0.6083)	0.0761 (0.3207)	0.2260 (<0.0001)	-0.0457 (0.6281)	-0.1331 (0.2476)
Distance to foveola	0.0074 (0.7155)	0.0012 (0.9038)	0.0357 (0.0119)	0.0040 (0.6398)	0.0149 (0.3822)	-0.0127 (0.5371)
Central macular thickness	-0.0005 (0.5203)	-0.0001 (0.7282)	0.0006 (0.2226)	<0.0001 (0.8519)	0.0002 (0.7234)	<0.0001 (0.9456)
Signal strength	0.0034 (0.5903)	-0.0008 (0.8067)	0.0085 (0.0483)	0.0007 (0.8003)	-0.0103 (0.0523)	-0.0047 (0.4546)

Multivariate analysis of clinical features associated with major OCTA artifacts. The *p*-values in bold and italics were considered statistically significant.

resulted is a greater number of certain artifacts, such as blink artifact, in even control eyes. Although it might be interesting to see how image quality improves with more scans taken, it is beyond our scope, as this study pertains to the image quality and frequency of image artifacts in normal versus diseased eyes. But, we did find that for our studies of the same study population by our group, most scans taken for foveal avascular zone and capillary density measurements were not from the first scan and rather the latter scans that have better quality. We acknowledge these limitations and realize there is no perfect way to determine frequency of artifacts, and our attempts to control variables may have increased certain artifacts in normal control eyes.

In summary, OCTA image artifacts are frequently seen in both normal and diseased eyes. However, image quality is overall better in nondiseased eyes with improved signal strength and fewer image artifacts. Ophthalmologists interpreting OCTA images should be familiar to these artifacts because analysis of capillary vascular density and FAZ may be affected. Future improvements in OCTA algorithms, eye tracking, and scan speeds could improve quality and lessen artifacts.

Key words: optical coherence tomography angiography, OCTA, signal strength, artifacts, segmentation, plaque radiotherapy, choroidal melanoma, uveal melanoma, radiation maculopathy.

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